SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I V Thanisha / 1965260160 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%				
Method Selection	20%				
Solution Process	25%				
Accuracy of Calculation	20%				
Interpretation of Results	15%				

Total Marks **/40** ***Signature of the Faculty***

ANALYTICAL PROBLEM SOLVING - QUESTIONS 1 TO 5

Question 1. 1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

Answer

Given and method

Original block: 192.168.20.0/24 (256 addresses). Lab A: /26; Lab B: /27; Lab C: /28. Assign subnets consecutively without overlap, starting at the first available network boundary.

Step 1: Calculate the subnet masks: /26 = 255.255.255.192; /27 = 255.255.255.224; /28 = 255.255.255.240.

Step 2: Formula: total addresses = 2^(32 - prefix); usable hosts = total addresses - 2 (network and broadcast).

Step 3: Lab A (/26): 2^6 = 64 total; 64 - 2 = 62 usable. Block size = 256 - 192 = 64. Network = 192.168.20.0; broadcast = 192.168.20.63; usable range = 192.168.20.1 to 192.168.20.62.

Step 4: Lab B (/27): 2^5 = 32 total; 32 - 2 = 30 usable. Next valid network = 192.168.20.64; broadcast = 192.168.20.95; usable range = 192.168.20.65 to 192.168.20.94.

Step 5: Lab C (/28): 2^4 = 16 total; 16 - 2 = 14 usable. Next valid network = 192.168.20.96; broadcast = 192.168.20.111; usable range = 192.168.20.97 to 192.168.20.110.

Final result

Lab A: 192.168.20.0/26 | Hosts .1-.62 | Broadcast .63 | 62 usable.

Lab B: 192.168.20.64/27 | Hosts .65-.94 | Broadcast .95 | 30 usable.

Lab C: 192.168.20.96/28 | Hosts .97-.110 | Broadcast .111 | 14 usable.

Interpretation: All three lab networks fit inside 192.168.20.0/24 without overlapping. Although the question calls these fixed subnet masks, the supplied /26, /27 and /28 prefixes are different sizes (VLSM).

Question 2. 2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

Answer

Given and method

System 1: 192.168.1.130; System 2: 192.168.1.190. Both use 255.255.255.192 = /26.

Step 1: Host bits = 32 - 26 = 6. Total addresses per subnet = 2^6 = 64. Usable hosts = 64 - 2 = 62.

Step 2: Block size in the fourth octet = 256 - 192 = 64. Subnet intervals are 0-63, 64-127, 128-191 and 192-255.

Step 3: System 1 has last octet 130. Since 128 <= 130 <= 191, it belongs to 192.168.1.128/26.

Step 4: System 1 network = 192.168.1.128; broadcast = 192.168.1.191; valid host range = 192.168.1.129 to 192.168.1.190. Thus .130 is a valid host.

Step 5: System 2 has last octet 190. Since 128 <= 190 <= 191, it also belongs to 192.168.1.128/26.

Step 6: System 2 network = 192.168.1.128; broadcast = 192.168.1.191; valid host range = 192.168.1.129 to 192.168.1.190. Thus .190 is also a valid host.

Final result

Both systems are in the SAME subnet: 192.168.1.128/26. Subnet range: 192.168.1.128-192.168.1.191. Valid hosts: 192.168.1.129-192.168.1.190. Usable hosts per subnet: 62.

Interpretation: Both IP addresses are usable and can communicate directly at the IP subnet level, provided their link and other settings permit it.

Question 3. 3. An organization has been allocated the IP block 172.16.0.0/16

Answer

Given and method

Network: 172.16.0.0/16. HR /25, Sales /26, IT /27, Admin /28. Allocate each department sequentially from the beginning, respecting subnet boundaries.

Step 1: Formula: total addresses = 2^(32 - prefix); usable hosts = total - 2.

Step 2: HR /25: mask 255.255.255.128; 2^7 = 128 total, 126 usable. Network 172.16.0.0; host range 172.16.0.1-172.16.0.126; broadcast 172.16.0.127.

Step 3: Sales /26: mask 255.255.255.192; 2^6 = 64 total, 62 usable. Network 172.16.0.128; host range 172.16.0.129-172.16.0.190; broadcast 172.16.0.191.

Step 4: IT /27: mask 255.255.255.224; 2^5 = 32 total, 30 usable. Network 172.16.0.192; host range 172.16.0.193-172.16.0.222; broadcast 172.16.0.223.

Step 5: Admin /28: mask 255.255.255.240; 2^4 = 16 total, 14 usable. Network 172.16.0.224; host range 172.16.0.225-172.16.0.238; broadcast 172.16.0.239.

Step 6: Total allocated addresses = 128 + 64 + 32 + 16 = 240. Total /16 addresses = 2^16 = 65,536. Unallocated addresses = 65,536 - 240 = 65,296.

Final result

HR: 172.16.0.0/25 (126 usable); Sales: 172.16.0.128/26 (62 usable); IT: 172.16.0.192/27 (30 usable); Admin: 172.16.0.224/28 (14 usable).

Unused address range: 172.16.0.240 through 172.16.255.255, containing 65,296 unallocated addresses. This is an address range, not one single CIDR subnet.

Interpretation: The four departments have non-overlapping allocations and substantial space remains for future departments.

Question 4. 4. A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address 172.16.10.95 falls within the same subnet range based on the given configuration.

Answer

Given and method

Original network: 172.16.10.0/24. New subnet mask: /27 = 255.255.255.224. Test IPs: 172.16.10.78 and 172.16.10.95.

Step 1: Borrowed bits = 27 - 24 = 3. Number of subnets = 2^3 = 8.

Step 2: Remaining host bits = 32 - 27 = 5. Addresses per subnet = 2^5 = 32. Usable hosts per subnet = 32 - 2 = 30.

Step 3: Block size = 256 - 224 = 32. Subnet starting octets: 0, 32, 64, 96, 128, 160, 192, 224.

Step 4: The /27 subnet ranges are 0-31, 32-63, 64-95, 96-127, 128-159, 160-191, 192-223 and 224-255.

Step 5: For IP 172.16.10.78, 64 <= 78 <= 95, so its network address is 172.16.10.64/27. Broadcast = 172.16.10.95. Valid hosts = 172.16.10.65 to 172.16.10.94.

Step 6: Check 172.16.10.95. It is inside the subnet address range 172.16.10.64-172.16.10.95, but is the BROADCAST address, not a usable host IP.

Final result

Total subnets: 8. Usable hosts per subnet: 30. IP .78 belongs to 172.16.10.64/27. Network: 172.16.10.64; broadcast: 172.16.10.95.

Interpretation: .95 belongs to the same subnet range as .78 but cannot be assigned to an ordinary host because it is reserved for broadcast.

Question 5. 5. A network uses the Internet Protocol version 6 address 2001:0db8:0000:0000:0000:ff00:0042:8329. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Answer

Given and method

IPv6 address: 2001:0db8:0000:0000:0000:ff00:0042:8329; prefix length: /64.

Step 1: An IPv6 address has 128 bits, written as eight groups of four hexadecimal digits (16 bits per group).

Step 2: Remove leading zeros from each group: 2001:db8:0:0:0:ff00:42:8329.

Step 3: Replace the longest consecutive run of all-zero groups with :: (only once). The shortest compressed address is 2001:db8::ff00:42:8329.

Step 4: To expand, replace :: with enough 0000 groups to restore eight groups and pad each group to four digits. Expanded address = 2001:0db8:0000:0000:0000:ff00:0042:8329.

Step 5: A /64 prefix means the first 64 bits (first four groups) identify the network; the final 64 bits identify the interface portion.

Step 6: Set the last four groups to zero to obtain the network prefix: 2001:0db8:0000:0000::/64, commonly written 2001:db8::/64.

Step 7: Host/interface bits = 128 - 64 = 64. Number of possible interface identifier combinations = 2^64 = 18,446,744,073,709,551,616.

Final result

Compressed IPv6: 2001:db8::ff00:42:8329.

Expanded IPv6: 2001:0db8:0000:0000:0000:ff00:0042:8329.

Network prefix: 2001:db8::/64. Theoretical address combinations in the /64: 18,446,744,073,709,551,616.

Interpretation: IPv6 does not use IPv4-style broadcast addresses; actual assignability can depend on reserved and special-use addresses. The /64 count is the mathematical address-space size.